FLOODCAST

Predict daily river discharge

A regression model using weather and basin characteristics, with an interactive river map for exploring observations and scenarios.

Public-data experiment · November 2026

Team member names pending confirmation

Three public data sources

312,363 daily records

55 stations · 2010-01-01 to 2025-12-31

USGS discharge	Observed daily mean river flow in cfs; actual station metadata.
Open-Meteo weather	Precipitation, min/max temperature and soil moisture from reanalysis.
USGS StreamStats	Published drainage area, basin elevation, forest cover and slope where available.

Daily mean is the public experiment target; confirm the event definition of “peak daily discharge.”

Weather, basin features, and time

Features

Rainfall and soil moisture, temperatures, physical basin attributes, seasonal terms, and preceding 3/7/14/30-day rainfall.

Models

Gradient boosting and ExtraTrees on raw, log, and drainage-normalized discharge. Imputation learns from training rows only.

Validation

Hold out later dates and complete stations. Freeze model selection before the separate 2025 holdout. No test discharge inputs.

Selected model: hist_log · 293,057 training rows

Measured regression performance

Evaluation	RMSE (cfs)	Peak RMSE*	MAE (cfs)	R^2
Later dates	56,184	60,396	18,706	0.832
Unseen stations	79,671	136,635	32,592	0.88
2025 holdout	52,695	113,268	16,064	0.891

Holdout station-median baseline RMSE: 85,500 cfs

*Extreme-event RMSE: actual discharge in the top 10% within each station. Public experiment results; not official competition scores.

Limitations and submission readiness

Flood peaks and unseen basins remain difficult

Gauge-point reanalysis does not represent every part of a large basin. Reservoir operations and incomplete basin attributes can limit accuracy.

A scenario is not a hydraulic forecast

The 3D flood overlay is an elevation-based exploration. It is separate from predicted discharge and omits channel hydraulics and flood defenses.

Before the official submission

Confirm the target and team names. Run against official test rows. Validate the generated CSV; do not edit predictions by hand.